

Quantifying Multimodal Review Quality with LLMs in the Online Marketplace

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Abstract— The contemporary AI ecosystem is rapidly expanding with a variety of LLM-based automated evaluation frameworks, including RAG evaluation tools, AI monitoring platforms, and text–image alignment evaluators. This study applies this emerging generative AI evaluation paradigm to e-commerce review analysis, examining how large language models can automatically assess review quality and how such AI-generated quality scores relate to actual marketplace performance. Specifically, we quantify multimodal review quality using three LLM-derived metrics: (1) text quality, (2) image quality, and (3) text–image alignment. We then analyze how these scores influence Amazon sales rankings. Machine learning classifiers showed that adding these LLM-based quality scores significantly improved sales-rank prediction. Hierarchical regression results show that, in addition to the individual contributions of text and image quality, incorporating cross-modal alignment substantially improves the explanatory power for sales rankings. These findings demonstrate the practical validity of LLM-based review evaluation mechanisms and offer an analytical framework that links multimodal review quality to real-world business outcomes.

Keywords—Large Language Models (LLMs); Review Quality; User-Generated Content (UGC); eWOM (Electronic Word of Mouth); Sales

I. ONLINE MARKETPLACE

The rapid growth of global retail e-commerce makes it increasingly important for stakeholders to understand the factors that drive marketplace performance. Prior research has shown that performance outcomes in digital marketplaces depend on product and service quality, user experience, and convenience [1], but also on the level of consumer trust.

A key mechanism for establishing trust is user-generated content (UGC)—particularly reviews—which supply additional information and have been empirically linked to sales performance [2]. On platforms such as Amazon, reviews that are voted to be “helpful” are algorithmically prioritized, directly shaping what consumers read and how they evaluate product credibility. Within this context, review quality affects consumer trust and product visibility, which in turn influence sales rank. [2], [3].

Companies like Uber Eats use generative AI to analyze and summarize key themes in text reviews, or enhances low

quality review images, showing the multimodal capacities of AI [4]. Retrieval-Augmented Generation (RAG) and Large Language Model (LLM) evaluation tools each use their own scoring methods and metrics. The underlying criteria are therefore heterogeneous, and it remains unclear which AI-derived metrics meaningfully relate to marketplace outcomes.

This study develops an explainable multimodal review quality framework that systematically integrates textual, visual, and text–image alignment dimensions. The core of this framework lies in decomposing ‘quality’ into dozens of fine-grained metrics, rather than treating it as a single, monolithic score. This approach enables an empirical validation of which specific AI-derived metrics meaningfully relate to marketplace outcomes. Using data from Amazon’s Best Seller rankings, we examine how LLM-derived metrics relate to sales rank and propose two applicable Review Quality Indices (RQI) for practical implementation. This research contributes to eWOM literature and the advancement of AI-based quality assessment by reconceptualizing review quality as an integrated multimodal construct.

II. LITERATURE REVIEW

A. Review Quality Metrics

Review quality refers to the informational and persuasive value of user-generated reviews that shape consumer trust and purchase intention. In textual reviews, product-specific and detailed information enhances consumer trust and decision quality [3], while readability and coherence improve persuasiveness [5]. Image-based reviews contribute additional informational cues—such as sharpness, lighting, and composition—that enhance authenticity and engagement [6]. Beyond individual modalities, text–image consistency, or the congruence between textual and visual content, strengthens review credibility and helpfulness by providing verifiable and coherent product information [6].

TABLE I. REVIEW QUALITY METRICS

Domain	Key Dimensions	References
Text	Relevance, Accuracy, Comprehensiveness, Content Richness & Specificity, Readability, Timeliness, Credibility & Authenticity, Organization	[3], [5], [7],[8]
Image	Technical Quality, Aesthetic Quality, Semantic Content, Affective / Implicit, Format & Context	[6],[9]
Alignment	Feature Verification, Modal Complementarity, Sensory Transfer, Contextual Replication, Affective–Consistency, Comparative Analysis	[6],[9]